

DHRUVIL BHAVSAR

Embedded | IoT | Process Automation | Industrial Control | Robotics

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PROFESSIONAL SUMMARY

Final-year Electronics & Instrumentation Engineering student with specialized expertise in **Embedded Firmware (C/C++)** and **IoT Hardware Architecture**. Proven track record of developing end-to-end prototypes, from PCB circuit design to firmware logic. Notable achievements include a **Gesture Control Robot featured in Gujarat Samachar** and a published research paper in IJERT. Seeking an embedded engineering internship to apply skills in ESP32, Sensor Interfacing, and Industrial Automation.

TECHNICAL SKILLS

- Microcontrollers:** ESP32 (Dual Core), Arduino (Uno/Nano), NodeMCU, STM32 (Basic).
- Firmware & Coding:** Embedded C, C++, Python (Scripting), Verilog.
- Protocols:** I2C (Inter-Integrated Circuit), SPI, UART, ESP-NOW (Low Latency WiFi), Bluetooth Classic/BLE.
- Sensors & Modules:** MPU6050 (6-Axis Gyro/Accel), AHT20/BMP280 (Environmental), DS3231 (RTC), DS18B20 (Temp), OLED (SSD1306), L293N (Motor Driver).
- Hardware Tools:** PCB Soldering & Troubleshooting, Multimeter Testing, Circuit Design (Fritzing), Power Supply Management.
- Software Tools:** Arduino IDE, VS Code, TIA Portal (PLC/SCADA), MATLAB.

FEATURED PROJECTS

1. IoT-Based Hand Gesture Control Robot | ESP32, MPU6050, C++ Featured in Gujarat Samachar & Published in IJERT Journal (Vol 12, Issue 09)

- Innovation:** Engineered a wireless robotic vehicle controlled via human hand gestures, designed to assist in contactless medicine delivery during health crises.
- Transmitter Logic:** Developed a wearable glove unit using **ESP32** interfaced with an **MPU6050 accelerometer** to calculate X/Y/Z-axis tilt angles in real-time.
- Communication:** Implemented **ESP-NOW protocol** to achieve low-latency (<10ms) data transmission between the glove and the rover, superior to standard Bluetooth range.
- Control System:** Programmed the receiver to map coordinate data into PWM signals for the **L293N motor driver**, enabling precise omnidirectional movement.

2. Smart IoT Mirror with Environmental Monitoring | ESP32, OLED, Sensors

- System Architecture:** Designed a smart reflective interface integrating an **OLED Display (SSD1306)**, **AHT20** (Temp/Humidity), and **BMP280** (Pressure).
- Firmware Optimization:** Interfaced multiple sensors on a single **I2C Bus**, resolving address conflicts and optimizing data polling intervals (2000ms) to prevent display flickering.
- Reliability:** Integrated a **DS3231 Real-Time Clock (RTC)** module to ensure timekeeping accuracy persisted even during network/power outages.

3. Smart Laptop Cooling System (Automation) | Arduino, Control Logic

- Automation Logic:** Built a closed-loop active cooling system using **Thermoelectric Cooling (Peltier TEC1-12706)** and water circulation.
- Sensor Fusion:** Programmed an Arduino UNO to read the **DS18B20** probe; devised a threshold logic where the system triggers a MOSFET/Relay only when temperature exceeds 40°C.
- Impact:** Achieved a temperature reduction of **8-10°C** during high-performance tasks while maintaining energy efficiency during idle states.

PROFESSIONAL EXPERIENCE

Industrial IoT Intern | Adbright Process Automation April 2025 – Present (Remote/Hybrid)

- Integrating IoT telemetry features into legacy **PLC-based automation systems**.
- Developing dashboards for real-time sensor data acquisition and system health monitoring to reduce downtime.

Robotics & Automation Intern | Indo German Tool Room (IGTR), Ahmedabad May 2023 – June 2023

- Programmed **4-axis and 6-axis robotic arms** for precise pick-and-place industrial operations.
- Gained practical exposure to **SCADA screen design** and Industrial PLC logic (Siemens TIA Portal).

EDUCATION

- B.E. Electronics & Instrumentation** | Vishwakarma Government Engineering College (VGEC)
 - Expected Graduation: 2026*
- Diploma in Instrumentation & Control** | Government Polytechnic Ahmedabad
 - Graduated: 2023*

CONTACT DETAILS

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- Languages:** English, Hindi, Gujarati (Native)

PERSONAL DETAILS

- Date of Birth:** 16/11/2004
- Gender:** Male
- Nationality:** Indian
- Availability:** Ready for Internship/Full-time roles starting April 2026